

Loan Approval - Prediction

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Abstract

This report explores the prediction of loan approvals using a tabular dataset containing 20,000 entries(rows) and 36 features(columns), including both categorical and numerical variables. The methodology incorporates comprehensive data preprocessing, feature selection, and the application of machine learning models. Key preprocessing steps included data cleaning, outlier removal, categorical encoding, and scaling. Feature selection was conducted using forward selection to optimize predictive accuracy. Machine learning models applied include K-Nearest Neighbors (KNN), Random Forest, and Logistic Regression, with the Random Forest model achieving the best accuracy. These results highlight the potential of advanced preprocessing and learning techniques in enhancing prediction outcomes. The findings have significant implications for financial institutions seeking to automate and refine loan approval processes.

1 Introduction

Loan approval processes are critical for financial institutions, impacting both organizational efficiency and customer satisfaction. Automating these processes with machine learning can reduce biases, streamline decisions, and improve accuracy. This study investigates the factors influencing loan approval decisions using a dataset of 20,000 applications, comprising attributes such as income, credit scores, and loan purposes.

The research focuses on understanding the relationships between features, identifying key predictors, and employing machine learning models to enhance decision-making. Initial analysis included data cleaning, encoding, and visualization to uncover trends and ensure dataset readiness. A correlation matrix and visualization techniques highlighted strong predictors.

The primary objective of this research is to build accurate predictive models, employing methods such as KNN, Random Forest, and Logistic Regression, and to identify the most effective approach. By doing so, the study aims to provide insights into how predictive analytics can optimize loan approvals.

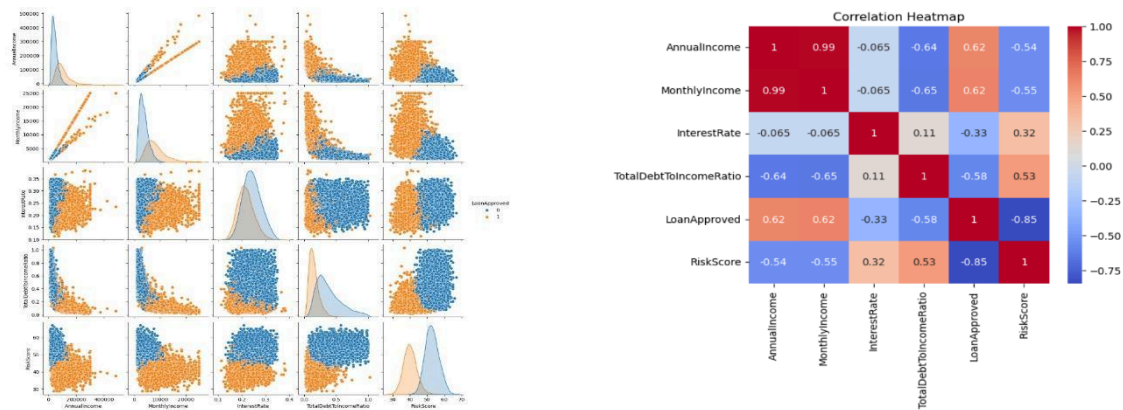
2 Methodology

Data Preprocessing

The dataset used in this study comprised 20,000 entries with 36 features, including both categorical and numerical variables. Preprocessing steps included data cleaning to ensure no missing values, followed by encoding categorical variables using one-hot encoding for multi-category features. Outliers were addressed using the Interquartile Range (IQR) method, ensuring a clean dataset free from extreme values. Additionally, features with low correlation with the target variable, LoanApproved, were identified and dropped. A correlation threshold of 0.3 (absolute) was set, meaning features with a correlation coefficient below this value were considered weak predictors and excluded from the final dataset to reduce dimensionality.

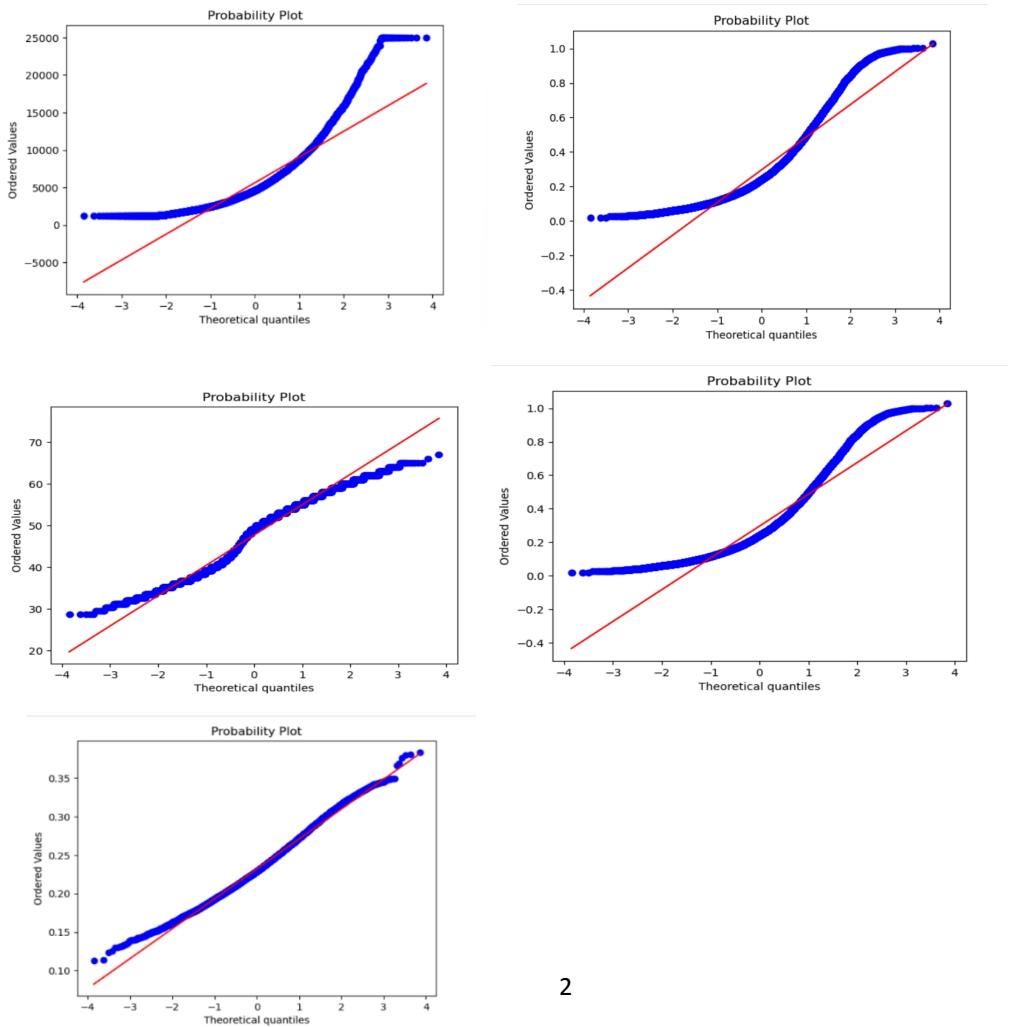
Data Visualization

I used Matplotlib and Seaborn to create heatmaps, pair plots, line plot and various other plots, helping us explore relationships between variables and understand the dataset better. These visualizations provided key insights about relation between different variables that became significant for feature selection.



Data Transformation

Data scaling was applied through Min-Max Scaling to standardize numerical features, aligning all variables within a consistent range for better model performance. Probability plots (probplots) were created to check if the data was normally distributed. As the data was not normally distributed, Min-Max Scaling was chosen over Z-score standardization to ensure better handling of the data's distribution.



One variable (InterestRate) was closely normally distributed, but I used Min-Max Scaling for it as well so that our numeric variables are all in same range.

3 Experiments

Machine Learning Models

Since loan approval prediction is a classification problem, I selected three machine learning algorithms to address it:

1. **Logistic Regression.**
2. **Random Forest.**
3. **K-Nearest Neighbors (KNN)** with k=3.

Feature Selection

To enhance model accuracy, forward selection was employed to identify the most significant features. This method added variables one by one based on how much they improved the model. If adding a variable reduced accuracy, it was not selected, ensuring the model used only the most important predictors.

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Feature Selection for Logistic Regression :
Added feature: RiskScore Accuracy: 0.9604
Added feature: AnnualIncome Accuracy: 0.9691
Added feature: InterestRate Accuracy: 0.9702
Added feature: TotalDebtToIncomeRatio Accuracy: 0.9767
Added feature: MonthlyIncome Accuracy: 0.9775

Selected Features for Logistic Regression : ['RiskScore', 'AnnualIncome', 'InterestRate', 'TotalDebtToIncomeRatio', 'MonthlyIncome']

Feature Selection for Random Forest :
Added feature: RiskScore Accuracy: 0.9919
Skipped feature: TotalDebtToIncomeRatio Accuracy did not improve.
Skipped feature: MonthlyIncome Accuracy did not improve.
Skipped feature: AnnualIncome Accuracy did not improve.
Skipped feature: InterestRate Accuracy did not improve.

Selected Features for Random Forest : ['RiskScore']

Feature Selection for K-Nearest Neighbors :
Added feature: RiskScore Accuracy: 0.8987
Added feature: AnnualIncome Accuracy: 0.9815
Skipped feature: MonthlyIncome Accuracy did not improve.
Skipped feature: TotalDebtToIncomeRatio Accuracy did not improve.
Skipped feature: InterestRate Accuracy did not improve.

Selected Features for K-Nearest Neighbors : ['RiskScore', 'AnnualIncome']
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4 Results & Discussion

Model Evaluation and Enhancement

Each model's performance was evaluated based on accuracy and classification report that includes precision, recall, f1score and macro and weighted average.

Classification Report for Logistic Regression :		precision	recall	f1-score	support
0	0.98	0.98	0.98	2129	
1	0.98	0.97	0.97	1434	

accuracy			0.98	3563
macro avg	0.98	0.98	0.98	3563
weighted avg	0.98	0.98	0.98	3563

Accuracy for Logistic Regression : 0.9775

Classification Report for Random Forest :		precision	recall	f1-score	support
0	0.99	0.99	0.99	2129	
1	0.99	0.99	0.99	1434	

accuracy			0.99	3563
macro avg	0.99	0.99	0.99	3563
weighted avg	0.99	0.99	0.99	3563

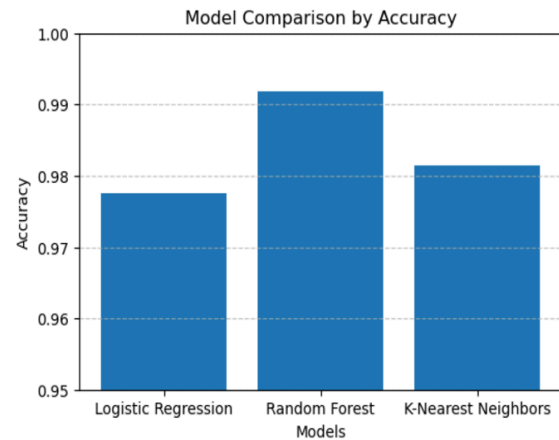
Accuracy for Random Forest : 0.9919

Classification Report for K-Nearest Neighbors :		precision	recall	f1-score	support
0	0.98	0.99	0.98	2129	
1	0.98	0.98	0.98	1434	

accuracy			0.98	3563
macro avg	0.98	0.98	0.98	3563
weighted avg	0.98	0.98	0.98	3563

Accuracy for K-Nearest Neighbors : 0.9815

Comparison of models



Among the algorithms tested, the **Random Forest model** yielded the highest accuracy and selected the least features to accomplish it, validating its effectiveness in handling the dataset's complexity.

Conclusion

The findings of this report demonstrate the potential of machine learning in automating loan approval processes, improving efficiency, and reducing biases. The Random Forest model, with its superior accuracy, proved to be a reliable tool for predicting loan approvals. This work highlights the importance of feature selection and preprocessing in enhancing model performance.